Herman Liang

Senior Full-Stack & Platform Engineer

Hermanyiqunliang@gmail.com — 437-340-7739 — Halifax, NS, Canada
linkedin.com/in/liang-yiqun — github.com/HermanL02 — hermanyiqunliang.com

Bridge Open Work Permit holder | no sponsorship required | PR application submitted

Professional Summary

Platform engineer who took a multi-tenant hospitality platform from zero to $1.1M+ in first-year gross booking value. Owning React/Next.js frontends, NestJS microservices, Kubernetes, event-driven pipelines, CI/CD, and observability end to end.

Experience

- **Suitely** — Remote, Canada
  Platform Software Engineer — ***July 2025 – Present***
  - Architected and led a TypeScript full-stack vacation rental platform (B2B/B2C) from zero to **$1.1M+** in first-year gross booking value — **4,970** confirmed reservations, **12,600+** room-nights, 200 properties, 484 room types.
  - Scaled channel connectivity to **8+** OTA/PMS integrations (Mews, DerbySoft, Hostex, Expedia, Booking.com, Google Hotels, Houfy, Airbnb), distributing **∼150K** date-level rate and availability updates daily across **545** channel listings with idempotent, retry-safe async processing.
  - Eliminated upstream rate-limit failures by designing a dedicated API gateway with scope-aware circuit breakers, interactive/bulk traffic lanes, and Redis-paced request quotas — **0 HTTP 429s across 315K+** third-party API calls since launch, down from a pre-gateway peak of 737 throttling failures/day.
  - Built the event-driven backbone on Google Pub/Sub — **10** topics with dedicated DLQs and **20** consumers — ingesting **45K+** webhook events/day and sustaining peaks of **152K** ARI messages/day on auto-scaled, multi-replica GKE services managed via Devtron.
  - Sustained **99.9%+** platform uptime and **p99 under 100ms** across **500K+** weekly requests by implementing an SRE-aligned stack (Sentry, OpenTelemetry, Pino, heartbeat monitoring, 20+ cron jobs).
  - Reduced **90%** of administrative workload by shipping AI tooling — an AI chat booking assistant, review insights, and conversation automation — alongside a fully automated Stripe booking and payment reconciliation pipeline.
  - Increased release confidence with a layered testing strategy (Vitest unit, Jest service-level, Playwright E2E) and zero-downtime additive migrations across 484 room types and **313** live pricing sync rules.

- **Koii Network (Koi Labs Inc.)** — Halifax, Canada
  Full Stack Developer — ***Mar 2024 – July 2025***
  - Developed AI-powered code analysis tools (RAG + LangChain) for semantic code understanding, improving search and comprehension across large repositories using a Flask server deployed on edge devices.
  - Led the full software development lifecycle for cross-platform products, including Electron desktop apps, MERN web applications, Chrome extensions, NPM packages, and CLI tools serving **100K+** users.
  - Integrated Solana programs with Node.js backends, abstracting blockchain complexity to allow developers to focus on core business logic.
  - Engineered Coolify infrastructure and assisted the DevOps team in configuring Devtron Kubernetes, enabling distributed task execution and seamless horizontal scaling across edge and cloud environments (AWS EC2, S3).
  - Designed and maintained GitHub Actions–based CI/CD pipelines with multi-stage workflows, enforcing test, lint, and security checks before production deployment, and enabling fast rollback to ensure deployment safety and system reliability.
  - Drove scalable P2P systems (Express.js, WebSockets) achieving **99.99%** uptime.

- **ALM Construction Group** — Remote
  Full Stack Developer — ***May 2023 – May 2024***
  - Spearheaded the design and development of comprehensive HR management and construction database systems (MERN+ stack), automating employee onboarding, attendance tracking, payroll, and construction data workflows for **300+ staff**, eliminating all manual processing and reducing operational errors by **95%**.
  - Revamped and modernized the legacy CMS (PHP), implementing containerized deployment using Docker on AWS EC2 instances, optimizing server resource utilization by 40% while ensuring **99.99% uptime** and seamless scalability.

Open Source & Technical Writing

- **Debugging a Silent Pub/Sub Stall in Production** — **Personal Blog**
  Node.js — Google Cloud Pub/Sub — Incident RCA — *Post*
  - Root-caused a silent production stall where subscribers stopped pulling under sustained load, tracing it to the Node.js client's default concurrency ceiling.
  - Published the incident timeline, debugging methodology, and fix — restoring steady-state throughput across event-driven services.
- **Koii Create Task CLI** — **834 downloads/week**
  Solana — NPM Package — *GitHub*
  - Enhanced a CLI tool enabling developers to create, manage, and deploy Koii Network tasks efficiently.
  - Integrated decentralized storage (IPFS) and managed KOII/KPL staking and reward distribution for tasks.
- **LiveSplitBoard** — **47 GitHub stars**
  Next.js — TypeScript — Splitwise API — *GitHub*
  - Built an open-source, mobile-first dashboard that tracks shared apartment expenses with roommates in real time via the Splitwise REST API.
  - Sends a Discord broadcast whenever a new expense is created, so roommates see updates without opening the app.

Education

- **MSc Computer Science | University of Warwick** — Coventry, UK
  Grade: Merit
- **BSc Computer Science | Dalhousie University** — Halifax, Canada
  Grade: Distinction with scholarship — Exchange @ City University of Hong Kong

Certifications

- **Cloud:** HashiCorp Certified: Terraform Associate (004), Microsoft Certified: Azure Fundamentals

Technical Expertise

- **Core:** TypeScript/Node.js, React/Next.js, NestJS, MongoDB, Redis, AWS, RAG
- **Architecture:** Event-driven microservices, Monorepo, CI/CD, Observability
- **Languages & Frameworks:** Python, Go, Express, Flask, Tailwind, React Native, Jest, Vitest, Playwright
- **DevOps/Cloud:** Docker, Kubernetes (GKE), Devtron, Terraform, GitHub Actions, Coolify; GCP, AWS (EC2, S3), Azure
- **Databases, MQ & APIs:** MongoDB, PostgreSQL, MySQL, Redis; Google Pub/Sub, Redis Streams; REST, GraphQL, gRPC